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CSE 445 Report

DASF-Net: Dual-Adversarial Spectral Fusion for Cross-Lingual and Adversarially Robust Audio Deepfake Detection

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Paper review 4 — WaveFake	Iftikhar Ahmed
Paper review 5 — Kawa <i>et al.</i> , adversarial defence	Meherun Nessa Mithila
Introduction, description of our own work	Iftikhar Ahmed
Proposed system (dataset and preprocessing)	Meherun Nessa Mithila
Proposed system (model description)	Iftikhar Ahmed
Testing methodology and evaluation protocol	Iftikhar Ahmed
Timeline and budget	Meherun Nessa Mithila
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Ten sections each; the two members contributed equally to this report.

DASF-Net: Dual-Adversarial Spectral Fusion for Cross-Lingual and Adversarially Robust Audio Deepfake Detection

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Abstract—Audio deepfake detectors are trained and evaluated almost exclusively on high-resource languages and on a narrow family of neural vocoders. When such a detector meets a new language synthesised by a structurally different generator, two failure modes compound: the learned features encode language- and corpus-specific cues rather than synthesis artifacts, and the decision boundary remains trivially breakable by gradient-based perturbation. This project proposes DASF-Net, a detection architecture built on the observation that these two failures are both *adversarial* problems and can be addressed by a single training formulation. DASF-Net couples (i) a dual-branch spectral front-end that presents the same utterance as a sparse cepstral view (LFCC) and a dense spectral view (log-Mel) to one Siamese-shared, ImageNet-pretrained Xception trunk; (ii) a Cross-Attention Spectral Fusion (CASF) module that lets the two views query one another and combines them through a learned gate, so that the network selects per utterance which representation carries the artifact; and (iii) a dual-adversarial objective in which a gradient-reversed language discriminator drives the embedding toward language invariance while adaptive FGSM/PGD adversarial training drives it toward perturbation robustness. A three-stage curriculum trains these parts in a stable order. The design is instantiated on the English WaveFake corpus (117,985 utterances, seven GAN- and flow-based vocoders) as source domain and the Bengali BanglaFake corpus (25,520 utterances, one VITS end-to-end system) as target domain, a pairing that varies language and synthesis paradigm simultaneously. A Confound-Controlled Evaluation Protocol is specified to neutralise the speaker, gender and class-prior imbalances that otherwise inflate Bengali detection scores. The expected impact is the first cross-lingual, adversarially evaluated benchmark for Bengali audio deepfake detection, together with a reusable methodology for extending it to other low-resource languages. This report covers the design, architecture, algorithms and experimental plan; the experiments have not yet been executed and no results are reported.

Index Terms—machine learning, audio deepfake detection, cross-lingual generalisation, adversarial robustness, domain-adversarial training, cross-attention fusion, Mel-spectrogram, Xception, convolutional neural networks, Bengali speech, low-resource language, equal error rate

I. INTRODUCTION, OBJECTIVES, MOTIVATION AND CONTRIBUTIONS

Neural vocoders and end-to-end text-to-speech (TTS) systems now produce speech that listeners routinely fail to separate from genuine recordings [1]–[5]. The resulting misuse — voice-impersonation fraud, fabricated evidence, political dis-

information — has driven a large body of work on automatic detection [6], [7]. That work is overwhelmingly English and is overwhelmingly built on a single family of generators.

Bengali, with more than 230 million native speakers, has almost no detection literature. BanglaFake [8] first made Bengali deepfake audio available at scale but trained no detector, and “Zero-Shot to Zero-Lies” [9] first benchmarked detectors on it, reporting a best zero-shot result of 53.80% accuracy (46.20% EER, Wav2Vec2-XLSR-53) and a best fine-tuned result of 79.17% accuracy (24.35% EER, ResNet18). Those numbers expose a gap that is not merely quantitative. A detector that transfers this poorly is not simply under-trained; it has learned the wrong thing — its features encode which corpus, which speaker and which language produced an utterance, rather than what a synthesiser did to it.

A. Motivation

The same detectors are, as a large body of adversarial work shows, breakable by perturbations far below the audible threshold. Kawa *et al.* [16] degrade an LCNN countermeasure from 0.022 to 0.785 EER under FGSM, and our group’s prior study [10] reproduces the pattern on a much larger ImageNet-pretrained Xception detector.

Our position is that these two failures share a structure. In both cases the network has latched onto a *nuisance direction* in feature space — language and corpus identity in one case, a locally exploitable gradient direction in the other. In both cases the standard remedy is to introduce an adversary that penalises exactly that direction. Domain-adversarial training [12] supplies the first adversary; FGSM/PGD adversarial training [14], [15] supplies the second. Placing both on a single shared embedding is, to our knowledge, new for audio deepfake detection, and it turns the cross-lingual question from a measurement exercise into an architectural one.

B. Objectives

The project pursues four measurable objectives.

- **Objective 1.** Design and implement DASF-Net, a dual-branch detector that fuses LFCC and log-Mel views through learned cross-attention over a weight-shared Xception trunk, and verify that the implementation is correct by shape and gradient tests.

- **Objective 2.** Establish an in-language Bengali baseline and determine whether the dense-over-sparse representation advantage reported for English CNN detectors [10] replicates on Bengali VITS audio, measured by Equal Error Rate (EER) against the published ResNet18 baseline of 0.2435 EER [9].
- **Objective 3.** Quantify zero-shot English $\leftrightarrow$ Bengali transfer with and without the language adversary, and report the reduction in EER attributable to domain-adversarial alignment.
- **Objective 4.** Measure whether adversarial robustness acquired in one language survives transfer to the other, reporting clean, FGSM and PGD EER on a single, unit-consistent ϵ grid.

C. Contributions

- 1) **DASF-Net**, an architecture that presents sparse cepstral and dense spectral views of one utterance to a single Siamese-shared Xception trunk and fuses them with a two-way cross-attention module and a learned gate.
- 2) A **dual-adversarial objective** combining a gradient-reversed language discriminator with adaptive FGSM/PGD adversarial training on the same fused embedding, plus a three-stage curriculum that makes the combination trainable.
- 3) A **Confound-Controlled Evaluation Protocol** (CCEP) for Bengali–English deepfake transfer, which neutralises the speaker, gender and class-prior artifacts that make naive BanglaFake accuracy figures uninterpretable.
- 4) A **corrected, unit-consistent adversarial threat model** that places FGSM and PGD on one shared ϵ grid, resolving an inconsistency present in our group’s own earlier work.
- 5) An **ablation-first experimental design**, so that any cross-lingual gain can be attributed to a specific component rather than merely observed.

II. PROJECT BACKGROUND

A. Context and Rationale

Speech synthesis has moved in a decade from concatenative systems with audible seams to neural systems whose output is, for untrained listeners, transparent. The BanglaFake authors report a Mean Opinion Score of 3.40 for naturalness and 4.01 for intelligibility on their Bengali synthetic speech [8], and their t-SNE visualisation of MFCC features shows the real and synthetic clusters overlapping heavily. In other words, the material is already good enough to deceive, and simple cepstral features are already insufficient to separate it.

The rationale for this project is that detection research has not kept pace in two specific directions. First, detectors are validated within one language, so nothing is known about their behaviour when deployed in another. Second, detectors are validated against passive forgeries rather than an active adversary who can compute gradients. A deployed countermeasure faces both conditions simultaneously.

B. Historical and Scientific Context

Anti-spoofing countermeasures grew out of the ASVspoof challenge series [6], where the dominant design was Linear Frequency Cepstral Coefficients (LFCC) with a Gaussian Mixture Model back-end. WaveFake [7] extended the setting to seven neural vocoders derived from LJSpeech [17], enabling leave-one-vocoder-out study of cross-generator generalisation, and reported that LFCC outperformed MFCC under a GMM classifier.

That ordering is back-end-specific. Our group’s prior work [10] showed that once the classifier becomes a deep convolutional network, a dense 128-bin log-Mel spectrogram transfers far better into an ImageNet-pretrained Xception CNN [11] than a sparse 20-coefficient cepstral representation, because the former retains the two-dimensional time–frequency structure that convolution and ImageNet initialisation both assume. The reported margin was a consistent 33–35% relative EER improvement across all seven WaveFake vocoders.

In parallel, domain-adversarial neural networks [12] established that a shared encoder can be forced to become uninformative about domain while remaining discriminative for the task, and adversarial training [14], [15] established the standard defence against gradient-based evasion. This project brings these two lines together for the first time in this application.

C. Significance to the Broader Domain

Roughly seven thousand languages are spoken worldwide and detection resources exist for a handful. A method that transfers a detector into a language with no labelled deepfake data — using only unlabelled target audio and a language tag — is worth considerably more than a method that requires a new labelled corpus per language. The design is not Bengali-specific: the language discriminator takes a language tag, and any low-resource language with unlabelled speech can be substituted. Demonstrating that adversarial robustness also transfers, or establishing that it does not, tells the field how much per-language work a secure deployment actually requires.

III. LITERATURE REVIEW

A. Existing Knowledge and Theoretical Basis

Four bodies of work form the foundation of this project.

Datasets and benchmarks. ASVspoof [6] standardised English and Chinese anti-spoofing evaluation. WaveFake [7] contributed 117,985 utterances across MelGAN, MelGAN-Large, Parallel WaveGAN, Multi-Band MelGAN, Full-Band MelGAN, HiFi-GAN and WaveGlow. BanglaFake [8] pairs 12,260 real Bengali utterances (SUST TTS Corpus [18], Mozilla Common Voice [19]) with 13,260 utterances from a VITS [5] system.

Feature representations. The LFCC-versus-Mel question, and the finding that the answer depends on the classifier family [7], [10].

Domain adaptation. The gradient reversal layer of Ganin *et al.* [12], which implements a minimax game between a feature

encoder and a domain classifier using a single backward-pass sign flip.

Adversarial machine learning. FGSM [14] and PGD [15] as white-box attacks, and adversarial training as the standard defence; attention [13] as the fusion mechanism used in our CASF module.

B. Gaps and Limitations in Previous Studies

- **BanglaFake [8] trains no detector.** It contributes data and a quality assessment only, so the dataset’s detection difficulty was never established by its authors.
- **Our prior work [10] is single-language and single-corpus.** It treats LFCC and log-Mel as mutually exclusive alternatives and explicitly lists hybrid representations as future work. It also downsamples to 16 kHz, discarding the 8–11 kHz band, and reports FGSM and PGD at ϵ values two orders of magnitude apart, which makes the two attacks non-comparable.
- **“Zero-Shot to Zero-Lies” [9] measures the gap but does not close it.** Its authors explicitly name cross-lingual generalisation and adversarial robustness as open problems for Bengali.
- **An unflagged confound in BanglaFake.** Its synthetic side is a single male speaker while its real side spans seven speakers of both genders, so a detector can score highly by classifying gender rather than synthesis artifacts. No published work on this dataset controls for it.
- **No cross-lingual adversarial study exists.** No prior work examines whether robustness learned in one language survives transfer to another.

C. Relevant Case Studies

Kawa *et al.* [16] provide the clearest case study of the adversarial failure mode, degrading an LCNN from 0.022 to 0.785 EER under FGSM and recovering most of it through adversarial training at a moderate clean-accuracy cost. Our prior study [10] reproduces the pattern at a much larger model scale, reducing FGSM error from 0.70 to 0.107 and PGD error from 0.90 to 0.154 with an adaptive attack-sampling scheme, at the cost of clean EER rising from 0.034 to 0.088. Together these establish both that the vulnerability is real across model scales and that adversarial training is an effective, if expensive, mitigation.

D. Relation and Advantage of This Project

This project sits directly on top of two of the works above and repairs the limitations of both. BanglaFake supplies the target-domain data it never used for detection; our prior Mel/Xception work supplies the source-domain method, which we extend rather than reuse.

Concretely, the advantages over the existing body of work are:

- 1) **Fusion instead of selection.** Where prior work chose between LFCC and log-Mel, CASF consumes both and learns a per-utterance gate — taking up the hybrid-representation direction that [10] left open, and yielding an interpretable output as a by-product.

- 2) **Closing the transfer gap rather than reporting it.** The gradient-reversed language head actively removes language information from the embedding, which the Zero-Shot to Zero-Lies benchmark identified as needed but did not attempt.
- 3) **Preserving the discriminative band.** Retaining 22.05 kHz keeps the 8–11 kHz region where vocoder and end-to-end TTS artifacts differ most, correcting a preprocessing choice inherited from our own earlier pipeline.
- 4) **A defensible evaluation.** CCEP controls the gender, speaker and class-prior confounds that make previously published BanglaFake accuracy figures difficult to interpret.
- 5) **A consistent threat model.** One ϵ grid for both attacks, correcting a unit inconsistency in our group’s prior paper.

IV. DESIGN METHODOLOGY AND PROPOSED SYSTEM

A. Overview of Approach

The project is carried out in four steps. First, both corpora are normalised to a common audio representation without discarding high-frequency content. Second, every utterance is expressed as two complementary spectral views and encoded by a single shared convolutional backbone. Third, the two views are fused by a learned, attention-based gate into one embedding, on which three prediction heads act. Fourth, that embedding is shaped by two adversaries at once — one local and perturbation-based, one global and domain-based — under a three-stage curriculum. Figure 1 shows the complete system.

B. Datasets

WaveFake (English, source domain) [7] contains 117,985 utterances (13,100 real, 104,885 synthetic) built from the single-speaker LJSpeech corpus [17], covering seven GAN-based or GAN-adjacent flow vocoders that resynthesise a waveform from acoustic features of real speech.

BanglaFake (Bengali, target domain) [8] contains 25,520 utterances: 12,260 real (10,000 from the SUST TTS Corpus, 2,260 from Common Voice across five speakers) and 13,260 synthesised by a single VITS model trained from scratch on the SUST corpus. VITS is an end-to-end variational/flow architecture that generates speech directly from text, and is therefore structurally unlike every WaveFake generator. Audio is distributed as 22,050 Hz WAV, 6–7 s per clip.

Because language and synthesis paradigm change together, the English $\leftrightarrow$ Bengali direction probes cross-lingual *and* cross-architecture generalisation at once. Table I summarises both corpora.

C. Data Preprocessing Pipeline

Every utterance from both corpora is brought to a common representation: mono, native 22.05 kHz *retained*, silences longer than 2 s trimmed at a 40 dB threshold, and duration normalised to 4 s by centre-cropping or reflection padding.

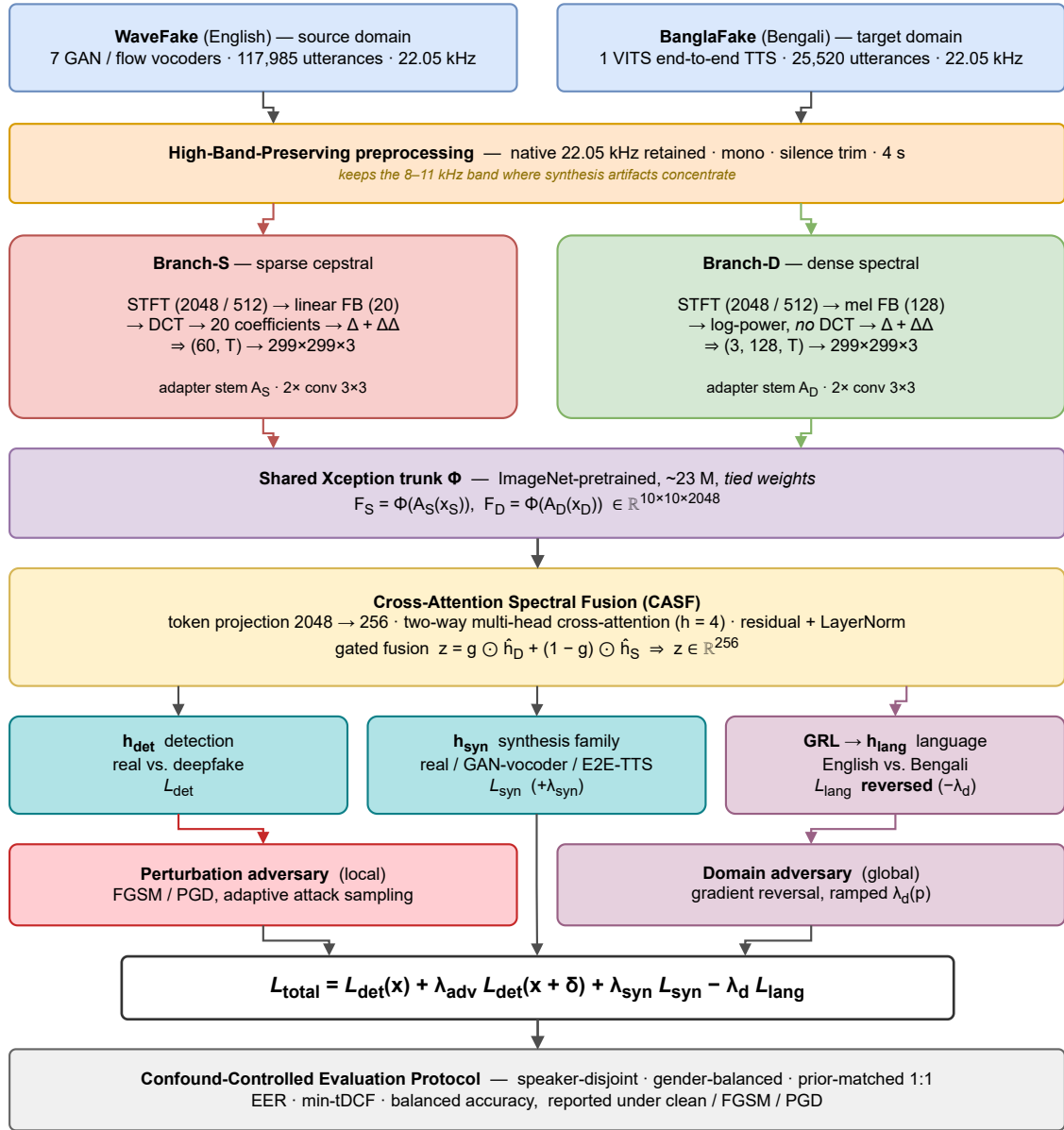


Fig. 1. System architecture and end-to-end pipeline of DASF-Net. Both corpora pass through identical High-Band-Preserving preprocessing, split into a sparse cepstral and a dense spectral branch, and are encoded by one Siamese-shared Xception trunk. Cross-Attention Spectral Fusion produces a single embedding z on which three heads act: detection, an auxiliary synthesis-family classifier, and a gradient-reversed language discriminator. Two adversaries — perturbation and domain — shape the same embedding through the combined objective, and all reported figures follow the Confound-Controlled Evaluation Protocol.

Retaining the native rate is a deliberate departure from our prior pipeline. Downsampling to 16 kHz imposes an 8 kHz Nyquist ceiling and removes the 8–11 kHz band; for a within-corpus study that loss is tolerable, but for a cross-paradigm claim it deletes the evidence most likely to distinguish a GAN vocoder from an end-to-end model. All filterbanks therefore run to $f_{\text{max}} = 11.025$ kHz. Every feature map is standardised per utterance and squashed to $[0, 1]$, so that the adversarial budget ϵ carries the same meaning for every utterance and for both branches.

Two parallel feature pipelines share identical Short-Time Fourier Transform parameters ($n_{\text{fft}} = 2048$, $\text{hop} = 512$), so any difference between them is attributable to the filterbank and compression stages alone:

- **Branch-S (sparse cepstral).** Linear-frequency filterbank (20 filters, 0–11.025 kHz) → Discrete Cosine Transform → 20 cepstral coefficients → first and second temporal derivatives, giving a $(60, T)$ representation.
- **Branch-D (dense spectral).** Mel-scaled filterbank ($n_{\text{mels}} = 128$, $f_{\text{max}} = 11.025$ kHz) → log-power

TABLE I
DATASET SUMMARY AFTER HIGH-BAND-PRESERVING ALIGNMENT

	WaveFake	BanglaFake
Role	source	target
Language	English	Bengali
Real utterances	13,100	12,260
Synthetic utterances	104,885	13,260
Synthesis family	7, GAN / flow	1, VITS end-to-end
Speaker(s), synthetic	1 female	1 male
Speaker(s), real	1 female	7 (mixed gender)
Source corpus	LJSpeech	SUST TTS Corpus
Sample rate (retained)	22,050 Hz	22,050 Hz

compression, with *no* DCT, so the dense time–frequency structure survives $\rightarrow$ first and second derivatives, giving a $(3, 128, T)$ tensor.

Both are resized to $299 \times 299 \times 3$, the Xception input geometry.

D. System Architecture

1) *Branch adapters and a Siamese-shared trunk*: Each branch owns a small *adapter stem* A_S , A_D — two 3×3 convolutions with batch normalisation and ReLU, about 0.06 M parameters each — which maps branch-specific input statistics into a common range before the shared encoder sees them. The encoder Φ is a single ImageNet-pretrained Xception [11] (≈ 23 M parameters) applied to both branches with tied weights:

$$F_S = \Phi(A_S(x_S)), \quad F_D = \Phi(A_D(x_D)), \quad (1)$$

with $F_S, F_D \in \mathbb{R}^{10 \times 10 \times 2048}$.

Weight sharing is a design commitment, not an economy. Two independent backbones would let each branch build a private feature space, and the fusion module would then have to reconcile two unrelated geometries. One trunk forces both views into a single artifact space at the cost of 0.12 M extra adapter parameters, and keeps the model comparable in size to the single-branch baseline of [10].

2) *Cross-Attention Spectral Fusion*: Each feature map is flattened into 100 spatial tokens and linearly projected to $d = 256$, giving $T_S, T_D \in \mathbb{R}^{100 \times 256}$. Two multi-head cross-attention blocks [13] ($h = 4$) let each view interrogate the other,

$$\hat{h}_D = \text{LN}(T_D + \text{MHA}(T_D, T_S, T_S)), \quad (2)$$

$$\hat{h}_S = \text{LN}(T_S + \text{MHA}(T_S, T_D, T_D)), \quad (3)$$

and a learned gate mixes them elementwise before mean-pooling over tokens:

$$g = \sigma(W_g [\hat{h}_D; \hat{h}_S] + b_g), \quad (4)$$

$$z = \text{pool}(g \odot \hat{h}_D + (1 - g) \odot \hat{h}_S) \in \mathbb{R}^{256}. \quad (5)$$

The gate is the point of the module. A fixed concatenation would assert that both views always matter equally; the gate instead lets the network decide, per utterance and per dimension, which representation carries the artifact. It also makes

the model interpretable: the distribution of g over a test set is a direct, quantitative answer to whether GAN vocoders and end-to-end TTS leave their traces in different representations.

3) *Objective heads*: Three heads read the same z . The **detection head** h_{det} ($256 \rightarrow 128 \rightarrow 2$) performs the task of record under cross-entropy with label smoothing 0.1. The **synthesis-family head** h_{syn} ($256 \rightarrow 128 \rightarrow 3$) predicts {real, GAN-vocoder, end-to-end TTS}; this supervision is deliberately coarse, because predicting the specific generator would invite memorisation of generator identity, which is exactly the failure that breaks cross-vocoder transfer. The **language head** h_{lang} predicts English versus Bengali but sits behind a gradient reversal layer (GRL) [12], which is the identity in the forward pass and multiplies the gradient by $-\lambda_d$ in the backward pass. The head therefore does its best to identify the language while the shared trunk and CASF are driven to make that identification impossible.

4) *The dual-adversarial objective*: Writing δ for an adversarial perturbation of the input features, the complete training objective is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{det}}(x) + \lambda_{\text{adv}} \mathcal{L}_{\text{det}}(x + \delta) + \lambda_{\text{syn}} \mathcal{L}_{\text{syn}} - \lambda_d \mathcal{L}_{\text{lang}}, \quad (6)$$

where the negative sign is realised by the GRL rather than by ascending the loss directly. The perturbation adversary is *local*: it searches a small neighbourhood of each input for a direction that flips the decision. The domain adversary is *global*: it searches the embedding for any direction that reveals the corpus. Our hypothesis is that they are complementary, and the ablation of Section V is designed to test that hypothesis rather than assume it.

E. Training Pipeline

Training equation (6) from random initialisation is unstable in practice: a gradient-reversed discriminator applied to an untrained encoder produces a large, uninformative reversed gradient, and adversarial examples generated against an untrained decision boundary are meaningless. Difficulty is therefore introduced in three stages (Fig. 3), each initialised from the previous.

Stage 1 — representation warm-up. Φ is frozen at its ImageNet weights; only A_S , A_D , CASF and h_{det} train, on labelled source data, under $\mathcal{L}_{\text{det}}$ alone. Adam, $lr = 10^{-4}$, effective batch 128 (micro-batch 16, accumulation 8), mixed precision.

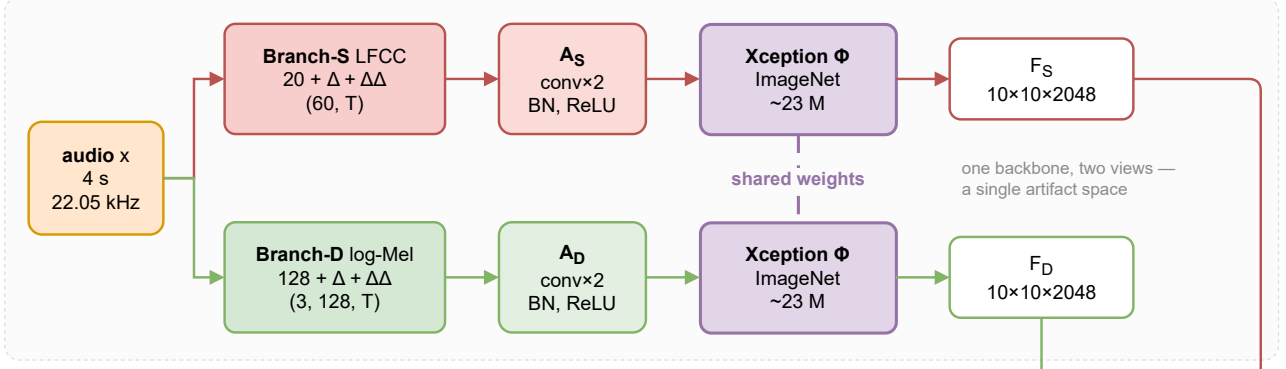
Stage 2 — cross-lingual alignment. Φ unfreezes; h_{syn} and h_{lang} are attached. Batches mix labelled source audio with *unlabelled* target audio: target labels are never used, only the language tag, which keeps the setting honestly unsupervised with respect to the target domain. Following [12] the reversal coefficient is ramped rather than fixed,

$$\lambda_d(p) = \lambda_{\text{max}} \left(\frac{2}{1 + e^{-10p}} - 1 \right), \quad (7)$$

with $p \in [0, 1]$ the fraction of the stage completed.

Stage 3 — dual-adversarial hardening. All modules fine-tune at $lr = 5 \times 10^{-5}$ for 10 epochs with adversarial examples,

(a) dual-branch encoder, Siamese weight sharing



(b) cross-attention spectral fusion and the three objective heads

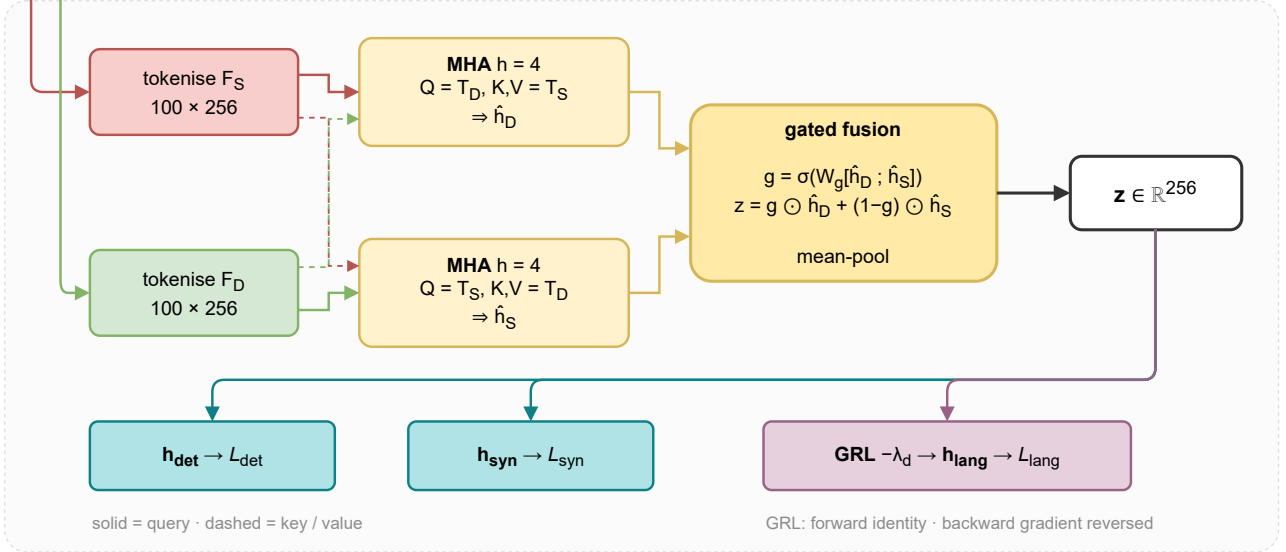


Fig. 2. DASF-Net in detail. (a) The dual-branch encoder: one utterance becomes a sparse cepstral and a dense spectral view, each passed through its own adapter stem into a single weight-shared Xception trunk. (b) CASF tokenises both feature maps, lets each view attend over the other, and combines them through a learned gate into z ; three heads then act on z , the language head behind a gradient reversal layer.

the language adversary still active. Rather than fixing the ratio of FGSM to PGD batches, the attack for each minibatch is sampled in proportion to a momentum-smoothed estimate of how successful each attack has recently been,

$$s_a \leftarrow (1 - m) s_a + m \hat{s}_a, \quad m = 0.2, \quad (8)$$

while a fixed fraction $p_c = 1/3$ of every minibatch stays clean to preserve baseline accuracy.

F. Testing Pipeline and Threat Model

Two white-box attacks are evaluated in the normalised feature domain, where inputs lie in $[0, 1]$. FGSM [14] takes a single step, $\delta = \epsilon \cdot \text{sign}(\nabla_x \mathcal{L}_{\text{det}})$, and PGD [15] iterates $K = 10$ steps with step size $\alpha = \epsilon/4$, projecting onto the ℓ_∞ ball of radius ϵ after each step. Critically, *both attacks are swept over the same ϵ grid*, $\epsilon \in \{0.001, 0.005, 0.010\}$. Prior

work, including our own, reports FGSM and PGD at ϵ values two orders of magnitude apart, which makes the resulting robustness figures non-comparable between the two attacks.

G. Tools and Resources

Table II lists the software and hardware used. All software is open source and both datasets are publicly available at no cost.

The 8 GB VRAM budget is a binding design constraint rather than an incidental detail. Because the shared trunk is evaluated twice per utterance, one sample occupies roughly double the activation memory of a single-branch Xception at 299×299 input, and a nominal batch of 128 does not fit. All training therefore uses a **micro-batch of 16 with gradient accumulation over 8 steps**, giving an effective batch of 128 and preserving the optimisation behaviour assumed by the learning rates above, at the cost of additional wall-clock time.

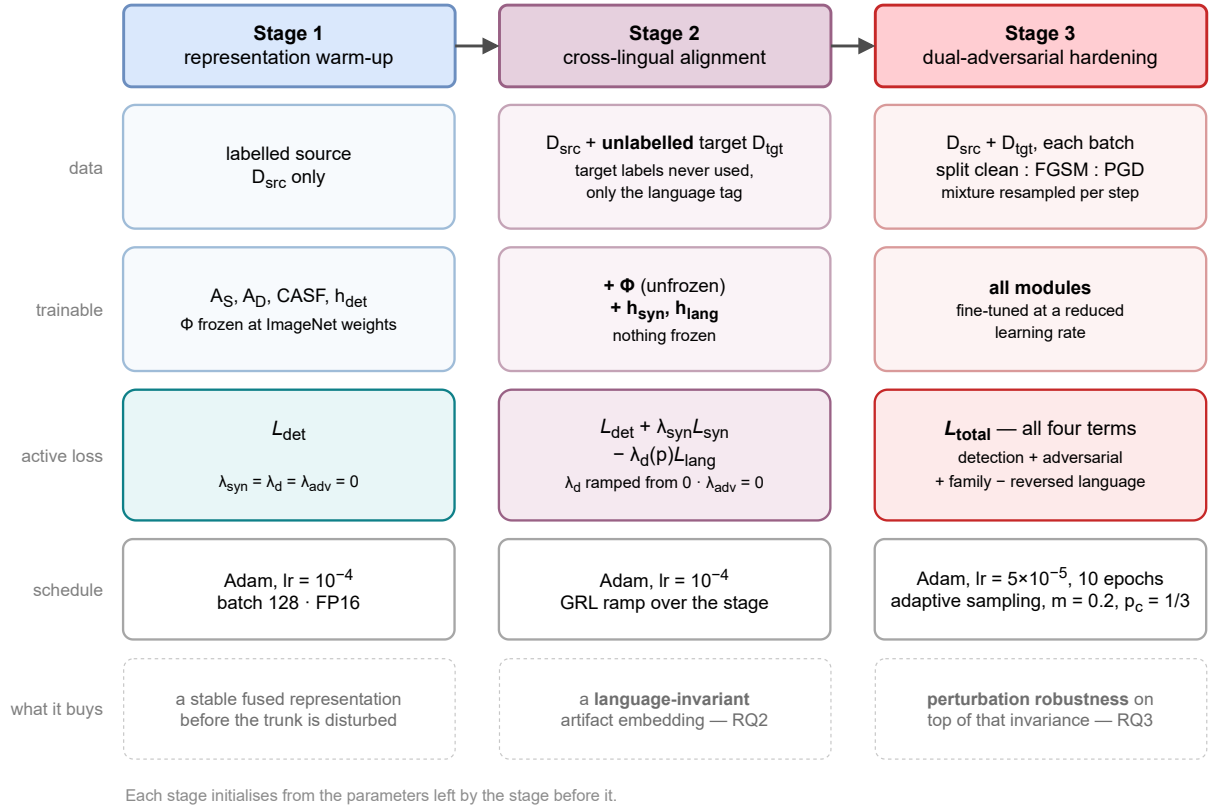


Fig. 3. The three-stage training curriculum. Each stage adds one source of difficulty and inherits the weights of the stage before it.

TABLE II
TOOLS, RESOURCES AND TECHNOLOGIES

Component	Choice
Language / framework	Python 3, PyTorch
Pretrained backbones	timm (ImageNet Xception)
Audio processing	librosa, soundfile, scipy
Metrics	scikit-learn (ROC / EER)
Data handling	numpy, pandas
Attacks	implemented directly in PyTorch autograd
Source dataset	WaveFake (public)
Target dataset	BanglaFake (HuggingFace, public)
Diagrams	draw.io
Typesetting	L ^A T _E X (IEEEtran)
Compute	NVIDIA RTX 5060 Ti, 8 GB (local)

Mixed precision (FP16) is used throughout, and adversarial examples are generated with the model parameters detached so that only input gradients are retained.

V. TESTING METHODOLOGY

A. How Success Will Be Measured

Success is defined against the four objectives of Section I, and is measured by the following criteria.

1) *Evaluation metrics*: The headline metric is **Equal Error Rate (EER)**, the operating point on the ROC curve where the false positive rate equals the false negative rate. EER is threshold-free, which matters here because the source corpus has a roughly 8:1 synthetic-to-real prior and the target corpus a roughly 1:1 prior; a threshold calibrated on one is miscalibrated on the other. **Balanced accuracy** is reported alongside it. Raw accuracy is never reported for a transfer condition without its matching prior.

The project’s evaluation plan also names min-tDCF, a standard anti-spoofing metric. It should be recorded explicitly that min-tDCF is defined over a *tandem* system — a counter-measure plus an automatic speaker verification subsystem — and that neither WaveFake nor BanglaFake ships ASV scores. It is therefore not computable from this data without adding an ASV subsystem, and is not reported.

2) *Benchmarks*: Three published reference points are used: the ResNet18 fine-tuned baseline on BanglaFake at 79.17% accuracy / 0.2435 EER [9]; the single-branch Xception with log-Mel at 0.034 average EER on WaveFake [10]; and the GMM+LFCC classical baseline at 0.058 EER [7].

3) *Confound-Controlled Evaluation Protocol*: Every reported figure is subject to four constraints, which follow from the confounds identified in Section III:

1) **Speaker-disjoint splits**. No speaker appears in both

TABLE III
PLANNED EXPERIMENTS AND THE OBJECTIVE EACH ADDRESSES

ID	Experiment	Obj.
E1	In-language baselines: Branch-S only, Branch-D only, and full CASF fusion, per corpus	2
E2	Zero-shot transfer in both directions, with and without the language adversary; per-vocoder breakdown; $\leq 10\%$ few-shot variant	3
E3	Clean, FGSM and PGD EER per ϵ , in-domain versus transferred robustness	4
E4	Ablation: remove CASF, shared trunk, h_{syn} , GRL, adversarial stage, or HBP, one at a time	1–4
E5	Gate analysis: distribution of g per synthesis family	1

training and test partitions, so a model cannot succeed by speaker recognition.

- 2) **Gender-balanced Bengali evaluation.** Because BanglaFake pairs one male synthetic speaker against mixed-gender real speech, the primary Bengali test set is restricted to male real speech, and the mixed-gender result is reported separately as a diagnostic. A large gap between the two is itself evidence of gender shortcutting.
- 3) **Prior-matched test sets.** Both test sets are resampled to 1:1 real-to-synthetic.
- 4) **Threshold-free primary metric,** as above.
- 4) *Experiments:* Five experiments are planned, summarised in Table III.

E4 is the experiment that converts an observed improvement into an attributed one. Without it, any gain remains unexplained; with it, the contribution of each component can be isolated. All experiments are run with three random seeds and reported as mean and standard deviation; single-seed results are not reported.

5) *Achievement of objectives:* Objective 1 is met when the implementation passes its correctness tests (tensor shapes, gradient-reversal sign flip, attack budget compliance). Objectives 2–4 are met when E1, E2 and E3 respectively produce seed-averaged EER figures under CCEP, whatever those figures turn out to be. A negative result — for instance, that the language adversary does not improve transfer — satisfies the objective provided it is measured under a protocol that could have detected an improvement.

B. Monitoring and Reporting

Progress is monitored through weekly supervisor meetings, a shared Git repository with all code and experiment configurations committed, and per-epoch training logs recording each loss term separately ($\mathcal{L}_{\text{det}}$, $\mathcal{L}_{\text{syn}}$, $\mathcal{L}_{\text{lang}}$) plus the running attack-success estimates. Recording the loss terms separately is what makes the GRL diagnosable: a language loss that collapses toward chance indicates the adversary is working, while one that collapses toward zero indicates it has overwhelmed the encoder.

TABLE IV
PROJECT TIMELINE AND MILESTONES

Phase	Activity	Status
1	Literature review; dataset acquisition and manifest construction <i>May–Jun 2026</i>	Complete
2	Architecture design; algorithm specification; diagrams; reference implementation <i>Jun–Aug 2026</i>	Complete
3	Report preparation and submission <i>Aug–6 Sep 2026</i>	Complete
4	Environment setup; training runs E1–E3 <i>from 6 Sep 2026</i>	In progress
5	Ablation E4 and gate analysis E5; results write-up <i>after Phase 4</i>	Pending

C. Current Status

No experiments have yet been executed. The architecture, the training algorithms, the threat model and the evaluation protocol are complete and the reference implementation is written, but it has not been trained, and this report therefore contains no results, no accuracy figures and no trained model. Reporting this plainly is preferred to presenting an untested design as a validated one.

VI. PROJECT TIMELINE

Table IV lists the phases and their status. Note that Phases 4 and 5 fall after the report deadline of 6 September 2026: the design and implementation were completed within the reporting period, but the training runs were not.

The critical-path risk is Phase 4: the dependency install and the first full training run must both succeed before Phase 5 can begin. Mitigation is described in Section IX.

VII. BUDGET

Training runs on a locally owned NVIDIA RTX 5060 Ti (8 GB), so the project incurs almost no direct monetary cost; the real cost is time and electricity. Table V gives the estimate. Compute demand is driven by two architectural decisions: the shared trunk runs twice per utterance, roughly doubling forward and backward cost relative to a single-branch model, and Stage 3 adversarial training with PGD at $K = 10$ multiplies the cost of every attacked minibatch. The estimate assumes on the order of 200 GPU-hours across all experiments and three seeds. The electricity and currency figures are order-of-magnitude estimates, assuming 1 USD $\approx$ 120 BDT, 8 BDT/kWh, and a sustained draw of about 0.2 kW.

The direct monetary cost is therefore negligible; the binding resource is the 8 GB VRAM ceiling and the wall-clock time that the resulting micro-batching implies. The contingency line is not part of the planned spend — it is the cost that would be incurred only if the local card cannot complete the runs within the available time, and it is listed so that the fallback is costed rather than assumed free.

TABLE V
ESTIMATED PROJECT BUDGET

Item	USD	BDT
Personnel (2 students, academic credit)	0	0
GPU hardware (RTX 5060 Ti, already owned)	0	0
Electricity, ≈ 200 GPU-h @ 0.2 kW	3	320
Storage, ≈ 150 GB local disk	0	0
Software and tools (all open source)	0	0
Datasets (both publicly released)	0	0
Total, direct cost	3	320
<i>Contingency: cloud fallback if local training proves infeasible (≈ 200 h)</i>		
	300	36,000

VIII. TEAM MEMBERS AND ROLES

The two student members contributed equally, taking ten report sections each as itemised in the Individual Contribution Table.

- **Iftikhar Ahmed** (2121019642) — model and method. Architecture design (CASF, the dual-adversarial objective, the three-stage curriculum), reference implementation, adversarial threat model, testing methodology, risk analysis, system diagrams, and $\text{\LaTeX}$ preparation of the report.
- **Meherun Nessa Mithila** (2221649042) — data and evaluation. Literature review and the five paper reviews, dataset acquisition and manifest construction, preprocessing pipeline, evaluation protocol, timeline and budget, and language editing.
- **Mohammad Abdul Qayum**, Assistant Professor, ECE Department — course faculty and supervisor. Scope definition, methodological review, and approval of the evaluation protocol.

A per-section breakdown of contributions is given in the Individual Contribution Table on the second page of this report.

IX. RISKS AND MITIGATION STRATEGIES

Risk 1 — Gradient reversal fails to converge. The GRL is a known-unstable component; an overly aggressive reversal coefficient can destroy the encoder’s discriminative capacity. *Mitigation:* the GRL is introduced only in Stage 2, never at initialisation, and λ_d is ramped from zero by the standard DANN schedule. The language loss is logged separately so divergence is visible immediately. If it still fails, the fallback is a non-adversarial alignment objective such as maximum mean discrepancy, and the failure is reported as a negative result rather than concealed.

Risk 2 — The gender confound inflates Bengali results. BanglaFake’s synthetic side is one male speaker against mixed-gender real speech, so a model can score highly by classifying gender. *Mitigation:* CCEP restricts the primary Bengali test set to male real speech and reports the mixed-gender figure separately as a diagnostic; a large gap between them is treated as evidence of shortcutting rather than success.

Risk 3 — Compute and memory ceiling. Training runs on a single 8 GB RTX 5060 Ti. The two-branch encoder roughly doubles both activation memory and training cost, and PGD at $K = 10$ multiplies the cost further; the full 117,985-utterance WaveFake corpus at three seeds is unlikely to fit in the available wall-clock time. This is the most probable cause of the project failing to produce results. *Mitigation:* the micro-batch of 16 with gradient accumulation over 8 steps keeps the effective batch at 128 within the VRAM ceiling. WaveFake is subsampled to a prior-matched subset, reducing 117,985 utterances to roughly 26,000 without changing the design — for this hardware, subsampling is a necessity rather than an option. PGD steps are reduced to $K = 5$ for ablation runs, with the full $K = 10$ budget reserved for the headline E3 numbers, and gradient checkpointing on the shared trunk is held in reserve if memory still binds.

Risk 4 — Only one Bengali generator exists. With a single VITS system, cross-vocoder generalisation within Bengali cannot be tested. *Mitigation:* claims are scoped explicitly to cross-*paradigm* transfer, and the limitation is stated in the report rather than left for a reviewer to find. Adding a Bengali FastSpeech2 or Tacotron2 system is identified as the next step.

Risk 5 — Schedule slip leaves the project without results. Phase 3 is on the critical path and depends on a successful environment setup. *Mitigation:* experiments are prioritised so that E1 and E2 — which answer Objectives 2 and 3 — run first; E4 and E5 are valuable but deferrable. A smoke test that exercises every code path on random tensors, without any dataset or pretrained download, allows the implementation to be validated before compute is committed.

Risk 6 — Environment and dependency failure. The project depends on PyTorch, timm and librosa, none of which is currently installed on the development machine, and the pretrained Xception weights require a download. *Mitigation:* the implementation accepts an injected backbone, so it can be validated with a lightweight stand-in trunk before the real dependency stack is installed.

X. CONCLUSION AND LIMITATIONS

This project proposes DASF-Net, an architecture that treats the two standing weaknesses of audio deepfake detectors — brittleness across languages and brittleness under perturbation — as instances of the same problem, and attacks both with adversaries acting on one shared embedding. It contributes a dual-branch spectral front-end over a Siamese-shared Xception trunk, a gated cross-attention fusion module that learns per utterance which spectral view carries the artifact, a combined objective in which a gradient-reversed language discriminator and adaptive FGSM/PGD training shape the same representation, a three-stage curriculum that makes this trainable, a corrected and unit-consistent adversarial threat model, and an evaluation protocol that neutralises the speaker, gender and prior imbalances which otherwise make Bengali detection results unreadable.

The work is positioned to be publishable because it addresses two problems that the most recent Bengali detection

benchmark [9] explicitly named as open, and because it does so with a mechanism rather than with more data or a larger model. If the central hypothesis holds, the result is a detector whose features describe what a synthesiser did rather than which corpus it was trained on — a property that generalises beyond Bengali to any low-resource language with unlabelled speech available.

A. Limitations

Several limitations bound what this project can conclude, and are stated in advance rather than discovered later.

- **No results yet.** The experiments of Section V have not been executed. Every claim in this report is a design claim, not an empirical one.
- **One Bengali generator.** BanglaFake contains a single VITS system, so the design tests cross-paradigm generalisation but cannot test cross-vocoder generalisation within Bengali the way WaveFake’s seven-generator protocol does for English. No such claim is made.
- **Confounds mitigated, not eliminated.** CCEP controls speaker, gender and prior, but each control shrinks the usable Bengali evaluation set, trading statistical power for validity.
- **White-box attacks only.** FGSM and PGD assume full gradient access. Black-box, transfer, perceptual and physical-channel attacks are outside this design.
- **Two languages.** Bengali and English differ in phoneme inventory, prosody and recording conditions simultaneously, so invariance demonstrated over one pair is evidence, not proof, of invariance in general.
- **Computational cost.** Stage 3 roughly doubles training time on top of the already doubled forward cost of the two-branch encoder.

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